

USV Surge Response

In part a) you should have come up with an equation of motion in the time-domain

$$M\dot{u}(t) + Du(t) = T(t)$$

b) Consider the scenario where the vessel starts at rest with no input. At time zero the input thrust is set to a constant 37.5 N.

We need to solve this first-order ODE with a step function forcing term: $T(t) = 37.5\mu(t)$ where $\mu(t)$ is the unit-step function. Taking the Laplace transfer function of the entire equation yields

$$M(sU(s) - u(0)) + DU(s) = \frac{37.5}{s}$$

We are told that the vessel starts at rest, so $u(0) = 0$. Solving for $U(s)$

$$U(s) = \frac{1}{Ms + D} \frac{37.5}{s}$$

To get back to the time-domain we need to take the inverse Laplace transform. If we do the PFE and then take the inverse Laplace transform we should get the expression

$$u(t) = 2(1 - e^{t/\tau})$$

We will learn that τ is the *time constant* of this first order response: $\tau = M/D = 1.92s$.

```
A = 37.5;
M = 36;
D = 18.75;
[r,p,k]=residue(A,[M, D, 0])

% Graph response
tt = linspace(0,15,100);
tau = M/D;
uu = (A/D)*(1-exp(-tt/tau));
figure(1)
clf()
plot(tt,uu)
grid on;
xlabel('Time [s]')
ylabel('Surge Velocity [m/s]')

% Find when 95% (you can do this by hand on the graph)
II = find(uu > 0.95*(A/D),1);
t95 = tt(II);
u95 = uu(II);
hold on
plot(t95,u95,'o','markerfacecolor','r')
legend('u(t)',sprintf('95%% at t = %.2f s',t95))
```

$r =$
-2
2
 $p =$
-0.5208
0
 $k =$
[]

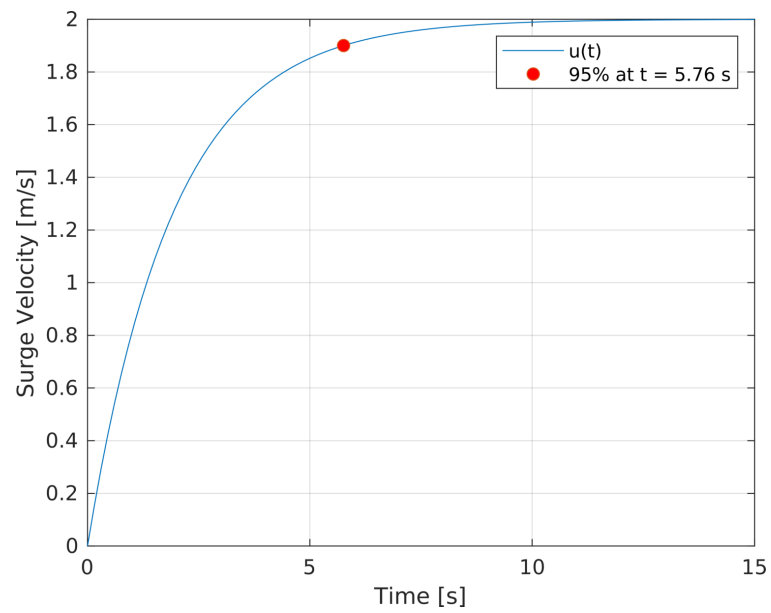


Figure 1: Graph of surge step-response.